Disease model

In their 1970s studies, Anderson and May focused first on building a mathematical model to show how a microorganism could affect a population. This led to a set of equations that they hoped would help explain the real-life impact of different kinds of pathogens, from bacteria and viruses to parasitic worms and insect larvae.

In their model, a number of mice were divided into three groups: susceptible (uninfected) mice, infected mice, and mice that had survived infection and were now immune. Unlike many earlier epidemiological models, the total population was not a fixed number; mice could be added either by reproduction or by additions from other populations. Mice also died from natural causes. In the absence of disease, the total would remain more or less the same, with the rate of added mice balancing that at which other mice died.

For simplicity, the model assumed that the diseases were transmitted by contact between infected and uninfected mice. Not all infected mice would die, so the model also included a recovery rate. Mice that recovered would be immune, at least initially. Immunity to viruses is more or less lifelong, but it is possible to become susceptible again to the same bacterial infection as time passes. Therefore, the calculations also included a rate of loss of immunity.

Putting all this together, Anderson and May produced a set of equations to predict the rate of population change in the three initial groups of uninfected but susceptible mice, infected mice, and the immune survivors. These equations could be added together to give the rate of change for the total mouse population.

From their calculations, they deduced that a disease will persist in a population whose equilibrium point (the rate of new additions, balanced by the natural death rate) is greater than the combined effects of natural